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Critical Evaluation and Optimization of the Thermodynamic Properties and Phase Diagrams of the CaO–FeO, CaO–MgO, CaO–MnO, FeO–MgO, FeO–MnO, and MgO–MnO Systems

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All available thermodynamic and phase diagram data at 1 bar pressure have been critically assessed for the liquid and solid (rock salt structure) phases of the six binary systems formed among the components CaO, FeO, MgO, and MnO. In the case of FeO-containing systems, data were evaluated at Fe saturation. All reliable data for each system were optimized simultaneously to give one set of model equations for the liquid and solid solutions. In systems for which the liquidus or solidus curves have not been measured, they were calculated. The thermodynamic properties and phase diagrams calculated from these equations are self-consistent and are the most reliable currently available estimates of the true values.

I. Introduction

THERMODYNAMIC and phase diagram data for a system should be assessed together in order that each can aid in the evaluation of the other. In a thermodynamic "optimization" of a binary system, all data are critically assessed simultaneously in order to obtain one set of model equations for the Gibbs energies of the phases as functions of temperature and composition. These equations then provide a consistent description of the phase diagram as well as the thermodynamic properties. They provide a consistent means of interpolation and extrapolation and of calculating phase boundaries which have not been measured.

Furthermore, the optimized parameters of the equations for the binary systems can then be used, along with appropriate models, in the prediction of the thermodynamic properties and phase diagrams of ternary and multicomponent systems. We are currently developing such an optimized database for the SiO_2–Al_2O_3–FeO–CaO–MgO–MnO–Na_2O–K_2O–ZrO_2–TiO_2–Ti_2O_3–S system. The present article reports on the evaluations of the six binary subsystems involving the components FeO, CaO, MgO, and MnO.

For the solid solutions in these six systems, excess molar Gibbs energies, G^E, are modeled by polynomial expansions in the component mole fractions. The liquid solutions, which are approximately regular, could be modeled in the same way. However, for other binary systems in the multicomponent database, particularly for those in which SiO_2 is a component, the liquid solutions are highly ordered, and polynomial expansions are totally inadequate for modeling the Gibbs energies. For such systems we have developed the modified quasi-chemical model[1–11] which has proved very successful. Since we are developing one database for the multicomponent liquid oxide solution, we must use the same model for all binary subsystems. Hence, the modified quasi-chemical model is used for the liquid phases in the present case. A brief outline of this model follows. A complete development has been given previously.[1,3] As shown below, when deviations from ideal solution behavior are small, the model approaches the polynomial expansion.

(1) *Modified Quasi-chemical Model*

In a molten oxide solution with components AO and BO we consider the formation of two second-nearest-neighbor A–B bonds from an A–A and a B–B bond:

$$(A - A) + (B - B) = 2(A - B) \tag{1}$$

The energy change for this reaction is represented by $\omega - \eta T$. Letting X_{AA}, X_{BB}, and X_{AB} be the fractions of each type of bond in solution, an equilibrium constant is obtained for reaction (1):

$$\frac{X_{AB}^2}{X_{AA}X_{BB}} = 4\exp\left(\frac{-2(\omega - \eta T)}{zRT}\right) \tag{2}$$

Two mass-balance equations may be written:

$$2X_{AO} = 2X_{AA} + X_{AB} \tag{3}$$

$$2X_{BO} = 2X_{BB} + X_{AB} \tag{4}$$

where X_{AO} and X_{BO} are the component mole fractions. The molar enthalpy, nonconfigurational entropy, and configurational entropy of mixing are given by

$$\Delta H = (b_{AO}X_{AO} + b_{BO}X_{BO})\left(\frac{X_{AB}}{2}\right)\omega \tag{5}$$

$$\Delta S^{nc} = (b_{AO}X_{AO} + b_{BO}X_{BO})\left(\frac{X_{AB}}{2}\right)\eta \tag{6}$$

$$\Delta S^{c} = -R(X_{AO}\ln X_{AO} + X_{BO}\ln X_{BO}) - \frac{Rz}{2}(b_{AO}X_{AO} + b_{BO}X_{BO})\left(X_{AA}\ln\frac{X_{AA}}{X_{AO}^2} + X_{BB}\ln\frac{X_{BB}}{X_{BO}^2} + X_{AB}\ln\frac{X_{AB}}{2X_{AO}X_{BO}}\right) \tag{7}$$

where R and T are the ideal gas constant and absolute temperature and where z, b_{AO}, and b_{BO} are constants which, as explained previously,[1,3] are set to $z = 2$, and $b_{AO} = b_{BO} = 0.6887$ for FeO, CaO, MgO, and MnO. Equations for the partial properties may also be written.[1,3]

Finally, ω and η are expanded as polynomials:

$$\omega = \omega_0 + \omega_1X_{BO} + \omega_2X_{BO}^2 + \omega_3X_{BO}^3 + \ldots \tag{8}$$

$$\eta = \eta_0 + \eta_1X_{BO} + \eta_2X_{BO}^2 + \eta_3X_{BO}^3 + \ldots \tag{9}$$

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The coefficients ω_i and η_i are the parameters of the model which are obtained by optimization of the data. A nonlinear least-squares optimization program has been written for this purpose.[5,12] Given a composition X_{BO} and values of the parameters ω_i and η_i, Eqs. (2) to (4) can be solved to give X_{AA}, X_{BB}, and X_{AB}, which can then be substituted into Eqs. (5) to (7) to give the thermodynamic properties.

If $\omega - \eta T = 0$, then, from Eqs. (2) to (4) it follows that $X_{AA} = X_{AO}^2$, $X_{BB} = X_{BO}^2$, and $X_{AB} = 2X_{AO}X_{BO}$. These are the values expected for an ideal (random) mixture. If $\omega - \eta T$ is small, then X_{AA}, X_{BB}, and X_{AB} are approximately equal to these ideal values. Substitution of these values into Eqs. (7) then shows that ΔS^c is nearly ideal, and substitution of Eqs. (8) and (9) along with $X_{AB} \approx 2X_AX_B$ into Eqs. (5) and (6) shows that Eqs. (5) and (6) reduce approximately to polynomial expansions.

(2) *Calculating the Liquidus from the Solidus*

In the CaO–MnO, MgO–MnO, and FeO–MgO systems, the components are completely miscible in the solid state at elevated temperatures. In each of these systems, the solidus, but not the liquidus, has been measured. Experimental activities in the solid solutions are also available. In previous publications[13,14] it has been shown that the liquidus, as well as the thermodynamic properties of the liquid, can be deduced, in a rigorous thermodynamic calculation, from a knowledge of the activities in the solid and the position and slope of the solidus, if the assumption is made that the entropy of mixing of the liquid is ideal. Such calculations were made in the present evaluations.

II. Gibbs Energies of Fusion of the Components

Here and elsewhere in this article, all temperatures have been converted to the International Temperature Scale of 1990.

(1) *CaO*

Much controversy surrounds the equilibrium melting point of CaO. Earlier measurements by several authors gave a melting point of 2595 ± 35°C. These data have been reviewed.[15-17] The melting point was remeasured in a solar furnace by Traverse and Foex,[15] who obtained 2927 ± 50°C. This value is preferred in the JANAF Tables. Recently a melting point of 2899°C was measured by digital pyrometry with an arc imaging furnace.[17] Traverse and Foex suggest that tungsten contamination from the sample supports was responsible for the much lower melting points observed in the earlier studies. However, these earlier studies were in good agreement with each other, and it seems unlikely that they would all suffer from exactly the same amount of contamination.

The solidus of the CaO–MnO system, which exhibits complete solid solubility, was measured by Schenck *et al.*[18] using an optical pyrometer over the composition range from 0 to 80 wt% CaO and between 1900° and 2400°C. The solidus clearly extrapolates to a melting point of CaO between 2550° and 2650°C as can be seen in Fig. 1. As discussed below, our thermodynamic assessment of this binary system, with an assumed melting point in this range, yielded simple expressions for the excess Gibbs energies of the solid and liquid solutions consistent with the measured solidus and with measured activities in the solid solution. The sample of Schenck *et al.* was not in contact with any container or support material which could have caused contamination. Furthermore, their measurements were made at temperatures below 2400°C where there are fewer experimental difficulties. The melting point obtained from extrapolation of their liquidus agrees very closely with that obtained by direct measurement in the earlier studies. Therefore, in the present assessments we have chosen a melting point in the temperature range of the early measurements. The chosen value of 2572°C is that given by Elliott and Gleiser.[19] As Ca has only one stable valence state, the effect of oxygen pressure on the melting point is expected to be very small.

The enthalpy of fusion of 79496 J/mol (19 kcal/mol) was taken from Kubaschewski *et al.*[20] The Gibbs energy of fusion is then

$$\Delta G^\circ_{\mathrm{fus(CaO)}} = 79496 - 27.9409T \text{ J/mol} \tag{10}$$

That is, $\Delta C_{p(\mathrm{fus})}$ has been set to zero. Thermodynamically, this amounts to one possible choice of standard state for the liquid below the melting point and for the solid above the melting point. It has been our experience, in the evaluation of a large number of systems, that this choice usually yields better results than are obtained when ΔC_p is calculated by extrapolating the heat capacity equations for the solid and liquid.

(2) *FeO*

A hypothetical congruent melting point of 1371°C[20] was adopted. The molar entropy of fusion of hypothetical solid stoichiometric FeO was assumed equal to that of $Fe_{0.947}O$ as reported by Robie *et al.*[21] since 1 mol of $Fe_{0.947}O$ contains 1 mol in total of cations and cation vacancies. $\Delta G^\circ_{\mathrm{fus}}$ was set to zero, such that

$$\Delta G^\circ_{\mathrm{fus(FeO)}} = 31189 - 18.9697T \text{ J/mol} \tag{11}$$

(3) *MnO*

The melting point of MnO can be strongly influenced by the presence of Mn^{3+}.[38] Early measurements of the melting point were up to 250°C lower than the most recently measured values, partly for this reason and probably partly because of the presence of other impurities. Schenck *et al.*,[18] as well as Fischer and Fleischer,[38] prepared MnO from the dissociation of $MnCO_3$ under H_2. The melting point was measured under high-purity Ar in a furnace which had been carefully purged of air. Melting points of 1878°C[38] and 1842°C[18] were reported. As discussed below in Section VI, there is reason to believe that the earlier[38] value is too high. Hence, the most recent value of 1842°C[18] has been accepted. This is also the value given in the recent compilation of Barin.[22] However, the enthalpy of fusion in the present evaluation was taken from the earlier compilation of Barin *et al.*[23] since in all cases this value permitted more consistent and simpler optimizations of many binary systems with MnO as a component. $\Delta C_{p(\mathrm{fus})}$ was set to zero.

$$\Delta G^\circ_{\mathrm{fus(MnO)}} = 54392 - 25.7154T \text{ J/mol} \tag{12}$$

(4) *MgO*

The melting temperature of 2825°C as well as $\Delta H^\circ_{\mathrm{fus}}$, $\Delta S^\circ_{\mathrm{fus}}$, and expressions for C_p(sol) and C_p(liq) were taken from Barin *et al.*[23] Above the melting point, $\Delta C_{p(\mathrm{fus})}$ was set to zero. Below the melting point, $\Delta C_{p(\mathrm{fus})}$ was calculated by extrapolation of the expression for C_p(liq). Since Mg has only one stable valence state, the effect of oxygen pressure on the melting point is expected to be very small.

$$\Delta G^\circ_{\mathrm{fus(MgO)}} = 77402 - 24.9852T \text{ J/mol} \quad (T > 2825°\mathrm{C}) \tag{13}$$

$$\Delta G^\circ_{\mathrm{fus(MgO)}} = 56647 + 70.8439T + (1.5711 \times 10^{-3})T^2 - 11.6859T \ln T - (0.5720 \times 10^6)T^{-1} \text{ J/mol} \quad (T < 2825°\mathrm{C}) \tag{14}$$

III. CaO–MnO System

The solidus was measured by Schenck *et al.*[18] from visual observation of the beginning of melting. Points taken from their smoothed curve over the composition range of the measurements are shown in Fig. 1. The preparation of MnO and the apparatus were the same as for the measurement of the melting point of MnO by the same authors as discussed in Section II.

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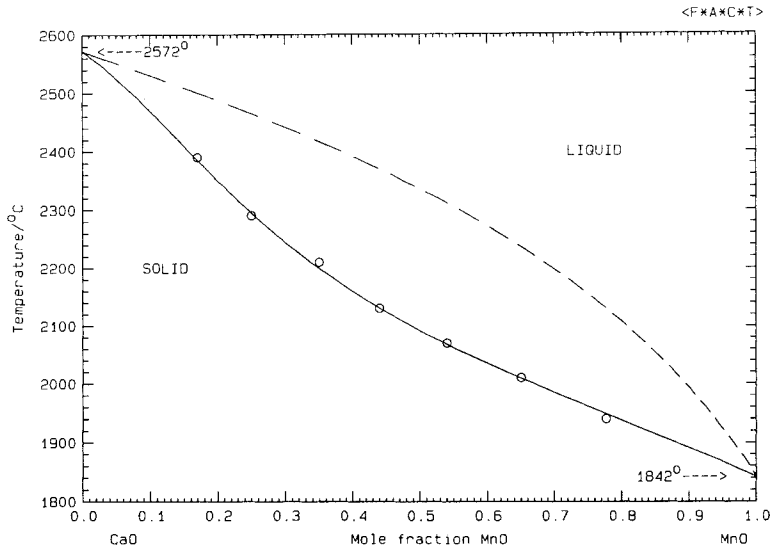

Fig. 1. Optimized CaO–MnO phase diagram. Experimental points from Schenck *et al.*[18]

From measurements of FeO activities in the ternary CaO–MnO–FeO system at 1100°C, Tiberg and Muan[24] showed that solid CaO–MnO solutions exhibit large positive deviations from ideality. CaO activities were measured by Raghavan *et al.*[25] using galvanic cells with CaF_2 solid electrolytes in the range from 940° to 1050°C. Their points are shown in Fig. 2.

The activities of Raghavan *et al.* were fitted to a regular solution model with

$$G^{E}(\text{sol}) = 15770 X_{CaO} X_{MnO} \text{ J/mol} \quad (15)$$

As can be seen in Fig. 2, the data are fitted quite well by this model, except at high X_{CaO}. However, the shape of the experimental activity curves in this composition range is inconsistent with Raoult's law. Hence, these experimental points were ignored. It is noteworthy that the temperature dependence of the activities is reproduced well by this model with only one temperature-independent parameter. A solid–solid miscibility gap with a consolute temperature at approximately 675°C is predicted.

Recently, activities of MnO in solid CaO–MnO solutions were reported,[71] based upon equilibrations with Pt-Mn alloys under controlled oxygen partial pressures, at 1500° and 1600°C. Agreement with Eq. (15) is good, except that the reported temperature dependence of the activities is very large, requiring an excess entropy of mixing which seems unreasonably large for a simple monoxide solution. Hence, these data were not used in the present assessment.

Given the solidus and the thermodynamic properties of the solid phase, the liquidus and the thermodynamic properties of the liquid phase were calculated thermodynamically as discussed in Section I. The resultant liquid properties were then represented by the quasi-chemical model with

$$\omega = 27006 \text{ J/mol} \quad (16)$$

The calculated phase diagram in Fig. 1 reproduces the measured solidus points very closely. Some calculated liquid thermodynamic properties are given in Table I.

The probable maximum inaccuracy in the assessed diagram is estimated as ±50°C for the solidus and ±100°C for the liquidus.

IV. MgO–MnO System

The solidus was measured by Schenck *et al.*[18] using a hot-stage microscope. Points shown in Fig. 3 are taken from the smoothed curve of these authors. The preparation of MnO and

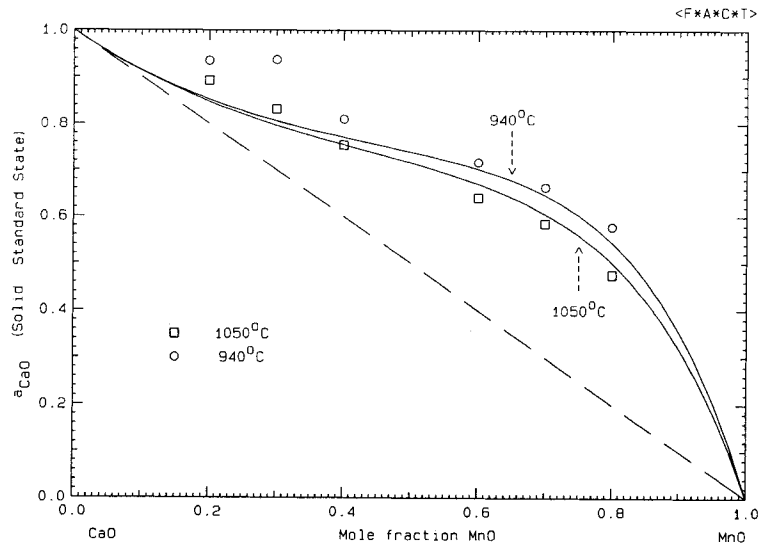

Fig. 2. Optimized activities of CaO in CaO–MnO solid solutions. Experimental points from Raghavan *et al.*[25]

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Table I. Calculated Partial Excess Gibbs Energies (Liquid Standard State) in Liquid Solutions*

System AO–BO	X_{BO}	$RT \ln \gamma_{AO}$ (J·mol^{-1})			$RT \ln \gamma_{BO}$ (J·mol^{-1})		
		2200°C	2400°C	2600°C	2200°C	2400°C	2600°C
CaO–MnO	0.0			0			18600
	0.1			253			13395
	0.2			840			10019
	0.3			1666			7527
	0.4		2690	2714		5500	5577
	0.5		3954	4001		3954	4001
	0.6		5500	5577		2690	2714
	0.7	7295	7419	7527	1652	1660	1666
	0.8	9725	9882	10019	848	844	840
	0.9	13093	13256	13395	263	258	253
	1.0	18600	18600	18600	0	0	0
		2200°C	2500°C	2800°C	2200°C	2500°C	2800°C
MgO–MnO	0.0			0			2882
	0.1			30			2301
	0.2			118			1802
	0.3			261			1374
	0.4		456	456		1008	1010
	0.5		704	706		704	706
	0.6		1008	1010		456	456
	0.7	1365	1370	1374	261	261	261
	0.8	1791	1797	1802	119	118	118
	0.9	2292	2297	2301	31	30	30
	1.0	2882	2882	2882	0	0	0
FeO–MgO	0.0	0	0	0	2306	2306	2306
	0.1	24	24	24	1841	1844	1846
	0.2	95	95	94	1442	1445	1448
	0.3	208	208	208	1099	1103	1105
	0.4		365	366		811	813
	0.5		566	567		566	567
	0.6		811	813		365	366
	0.7			1105			208
	0.8			1448			94
	0.9			1846			24
	1.0			2306			0
		2500°C	2700°C	2900°C	2500°C	2700°C	2900°C
MgO–CaO	0.0			0			31219
	0.1			583			18402
	0.2		1711	1722		11619	11789
	0.3		3096	3133		7417	7507
	0.4		4638	4705		4531	4566
	0.5	6148	6255	6348	2540	2542	2543
	0.6	7725	7851	7962	1240	1225	1212
	0.7	9165	9294	9408	456	439	424
	0.8	10278	10388	10485	76	65	55
	0.9		10830	10886		−23	−26
	1.0		10157	10157		0	0
		1400°C	1600°C	1800°C	1400°C	1600°C	1800°C
CaO–FeO	0.6		−14800	−12269		−5965	−5667
	0.7	−21923	−18600	−15574	−3817	−3877	−3859
	0.8	−25594	−22338	−19210	−2612	−2651	−2669
	0.9	−34470	−31375	−28311	−1143	−1154	−1163
	1.0	−60202	−57320	−54439	0	0	0

*Note: FeO–MnO ideal.

the apparatus were the same as for the measurement of the melting point of MnO by the same authors as discussed in Section II.

Activities of MgO in solid solutions in the range 890° to 1045°C were measured by Raghavan *et al.*[26] using galvanic cells with MgF_2 solid electrolytes. Their results are plotted in Fig. 4. Calculations of MgO activities were performed by Muan and co-workers[27,28] based upon a study of the MgO–MnO–NiO ternary system at 1100°C. These calculations indicate larger positive deviations than those reported by Raghavan *et al.* In the present analysis, the data of Raghavan *et al.* were used since these were directly measured for the binary system and since they yield a very simple analysis for the system as discussed below.

The activities of MgO in the solid solution were represented by a regular solution model:

$$G^E(\text{solid}) = 9252 X_{MgO} X_{MnO} \text{ J/mol} \quad (17)$$

As can be seen in Fig. 4, this gives very good agreement with the measured activities, even as regards the temperature dependence.

Recently, activities of MnO in solid MgO–MnO solutions were reported,[71] based upon equilibrations with Pt-Mn alloys under controlled oxygen partial pressures, at 1500° and 1600°C. Agreement with Eq. (17) is good, except that the reported temperature dependence of the activities is very large, requiring an excess entropy of mixing which seems unreasonably large for a simple monoxide solution. Hence, these data were not used in the present assessment.

Given the solidus and the thermodynamic properties of the solid phase, the liquidus and the thermodynamic properties of

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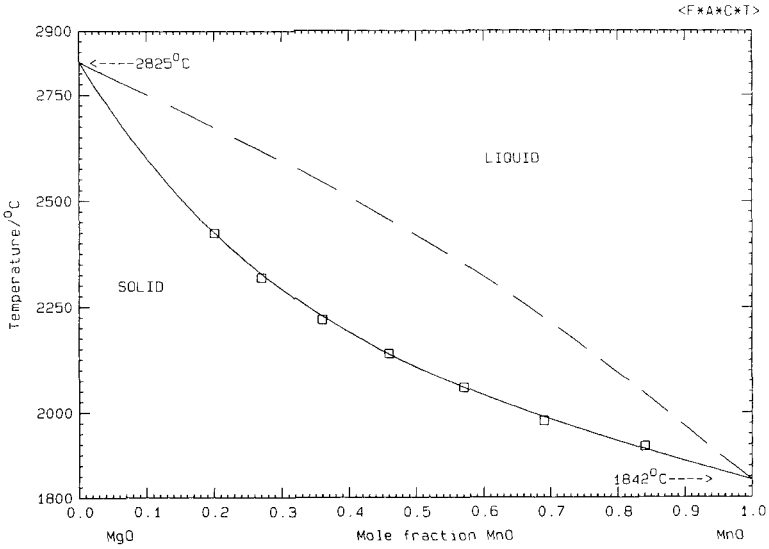

Fig. 3. Optimized MgO–MnO phase diagram. Experimental points from Schenck *et al.*[18]

the liquid phase were calculated thermodynamically as discussed in Section I. The resulting liquid properties were then represented by the quasi-chemical model with

$$\omega = 4184 \text{ J/mol} \tag{18}$$

Some calculated liquid thermodynamic properties are given in Table I. Positive deviations from ideality are greater in the solid solution than in the liquid solution. This is consistent with usual behavior. The calculated phase diagram is shown in Fig. 3. The solidus measurements are well reproduced.

The enthalpy of mixing of solid solutions was measured at 40°C by Gripenberg *et al.*,[29] who obtained $\Delta H(\text{sol}) \approx 20000 X_{MgO} X_{MnO}$ J/mol. Combination of this with Eq. (17) for G^E at ~950°C would give an excess entropy of $S^E(\text{sol}) \approx 8.8 X_{MgO} X_{MnO}$ J/(mol·K). However, in view of the 1000° temperature difference between the measurements of Raghavan *et al.* and Gripenberg *et al.* and in view of the uncertainties in both these studies, it was assumed in the present analysis that G^E of Eq. (17) is independent of temperature.

The probable maximum inaccuracy in the assessed diagram is estimated as ±50°C for the solidus and ±100°C for the liquidus.

V. FeO–MgO System

The assessment of this system has already been presented elsewhere.[11] However, it is repeated here for the sake of completeness. The phase diagram shown in Fig. 5 is for the system in equilibrium with metallic Fe with all Fe^{3+} converted to "FeO." That is, the composition axis is to be interpreted as $X_{MgO} = n_{Mg}/(n_{Mg} + n_{Fe^{2+}} + n_{Fe^{3+}})$ where n_i is the number of moles of i, and the diagram is at unit Fe activity.

It was first shown by Bowen and Schairer[30] that solid FeO and MgO are completely miscible at high temperatures. Solidus temperatures were determined by Schenck and Pfaff[31] up to 2400°C by a visual technique.

Several thermodynamic investigations of the solid solutions[32–36] have been reviewed.[37] Results are in poor agreement. Furthermore, the nonlinear variation of Fe^{3+} content with composition was often neglected in the derivation of FeO activities. A reinvestigation of oxygen potentials of FeO–MgO solid solutions equilibrated with Fe was made by Srecec *et al.*[37] between 1060° and 1320°C. Fe^{3+} contents varied from ~10% of total Fe for pure FeO to ~2% at $X_{MgO} = 0.5$. As shown in Fig. 6, the FeO activities of these authors are well represented by a regular solution model with

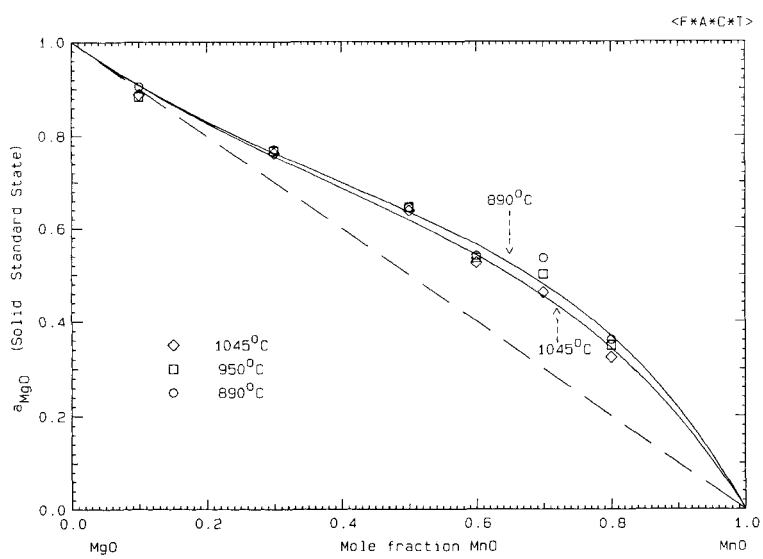

Fig. 4. Optimized activities of MgO in solid MgO–MnO solutions. Experimental points from Raghavan *et al.*[26]

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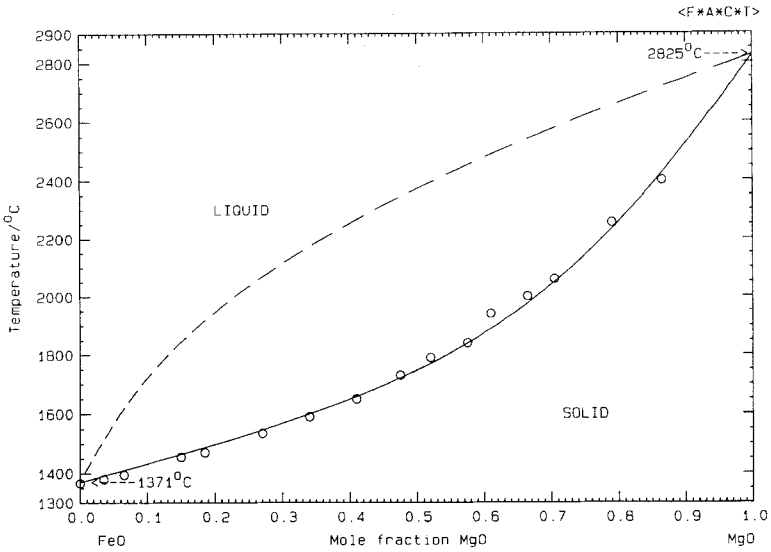

Fig. 5. Optimized FeO–MgO phase diagram for equilibrium with Fe. Experimental points from Schenck and Pfaff.[31]

$$G^{E}(\text{sol}) = 10175 X_{FeO} X_{MgO} \text{ J/mol} \quad (19)$$

These "FeO" activities are to be interpreted as

$$a_{FeO} = (P_{O_2}/P^{\circ}_{O_2})^{1/2} \quad (20)$$

where P_{O_2} is the equilibrium oxygen pressure over the Fe-saturated slag and $P^{\circ}_{O_2}$ is the value of P_{O_2} when $X_{FeO} = 1$.

Given the solidus and the thermodynamic properties of the solid phase, the liquidus and the thermodynamic properties of the liquid phase were calculated thermodynamically as discussed in Section I. The resulting liquid properties were then represented by the quasi-chemical model with

$$\omega = 3347 \text{ J/mol} \quad (21)$$

Some calculated liquid thermodynamic properties are given in Table I. The calculated solidus in Fig. 5 is in very good agreement with the measurements.[31] Positive deviations from ideality are smaller in the liquid than in the solid, as is consistent with normal behavior. The calculated consolute temperature of the solid–solid miscibility gap is ~340°C.

The probable maximum inaccuracy in the assessed diagram is estimated as ±50°C for the solidus and ±100°C for the liquidus.

VI. FeO–MnO System

The composition axis of the phase diagram (Fig. 7) for the system in equilibrium with Fe is to be interpreted as discussed above in Section V.

Liquidus points measured by Fischer and Fleischer,[38] by thermal analysis, are shown in Fig. 7. These authors reported melting points of FeO and MnO of 1372° and 1878°C, respectively. The latter is 36° higher than the presently assessed value. In the experiments, Fe crucibles were used for samples with less than 40 mol% MnO, and MnO crucibles were used for samples with higher MnO contents. The reported liquidus exhibited a marked kink near 40 mol% MnO. Hence, in Fig. 7, all liquidus points obtained in MnO crucibles have been adjusted downward by $36X_{MnO}$ degrees. The resultant curve is smooth. (For clarity, only a few of the 35 reported liquidus points are shown in Fig. 7. These span the reported composition range.)

The nonstoichiometry of FeO–MnO solid solutions has been studied by Engell and Kohl[39] at two selected oxygen pressures at 1250°C. The trivalent cation content of solid solutions equilibrated with Fe at high temperatures is of the order of 10% for pure FeO, 3% when $X_{MnO} = 0.5$, and <0.5% for pure MnO. The oxygen pressures of FeO–MnO solid solutions in equilibrium with Fe have been determined in several studies.[40–42] In the

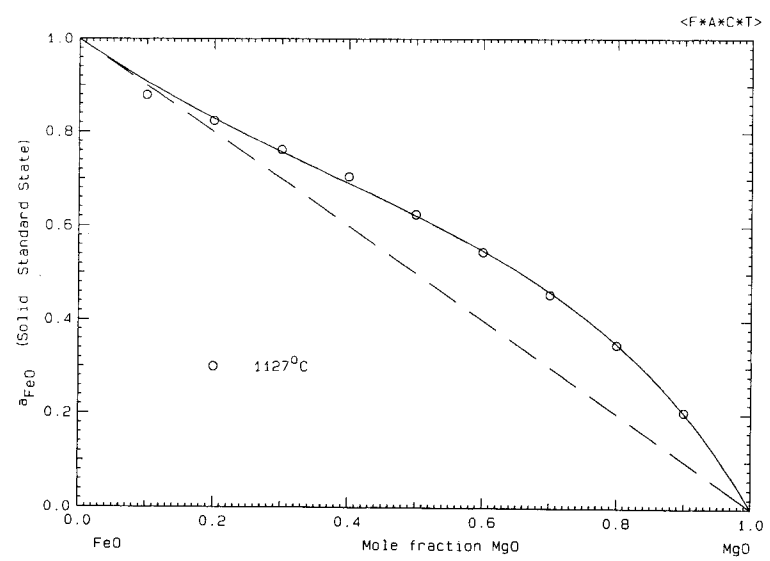

Fig. 6. Optimized FeO activities in FeO–MgO solid solution at 1127°C in equilibrium with Fe. Experimental points from Srecec *et al.*[37]

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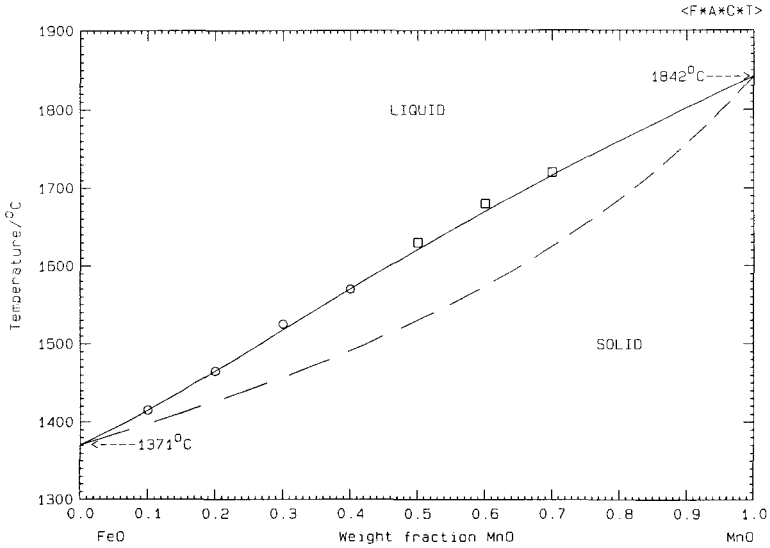

Fig. 7. Optimized FeO–MnO phase diagram for equilibrium with Fe. Experimental points from Fischer and Fleischer:[38] (○) unadjusted, (□) adjusted.

The oxygen pressures of FeO–MnO solid solutions in equilibrium with Fe have been determined in several studies.[40–42] In the two former studies the condensed phases were equilibrated with CO/CO_2 mixtures, whereas in the latter study a galvanic cell with a calcia-stabilized zirconia solid electrolyte was used. The results of these studies are not in good agreement. The FeO activities of Foster and Welch[41] are nearly ideal, those of Engell[42] show pronounced positive deviations, and those of Schenck *et al.*[40] show moderate positive deviations.

More recently, oxygen pressures above FeO–MnO solid solutions in equilibrium with Fe were measured by Schwerdtfeger and Muan[43] at 1000°, 1150°, and 1300°C. No temperature dependence was observed. FeO activities taken from the smoothed curve are shown in Fig. 8 (where a_{FeO} is defined as in Eq. (20)). Small positive deviations were observed, in reasonable agreement with the results of Schenck *et al.*[40] Activities of FeO–MnO solutions in equilibrium with Fe in the temperature range 800° to 1100°C were reported by Abbattista and Maja.[44] At 1100°C their results are in good agreement with those of Schwerdtfeger and Muan. However, a fairly large temperature dependence was reported. For a solution exhibiting nearly ideal behavior, the nearly zero temperature dependence reported by Schwerdtfeger and Muan seems more likely. Accordingly, the activity measurements of these authors were fitted in the present analysis to a regular solution model (See Fig. 8) with

$$G^E(\text{sol}) = 6126 X_{FeO} X_{MnO} \text{ J/mol} \tag{22}$$

The liquid was assumed to be ideal. The calculated liquidus in Fig. 7 agrees very well with the measured[38] points.

The probable maximum inaccuracy in the assessed diagram is estimated as ±25°C for the liquidus and ±50°C for the solidus.

VII. MgO–CaO System

The optimized phase diagram is shown in Fig. 9. Early reports of the eutectic temperature and eutectic liquid composition include 2300 ± 50°C at 59 mol% CaO,[45] 2360°C at 55 mol% CaO,[46] and 2250°C at 55 mol% CaO.[47] Based on considerations of lattice parameters and the observed solid solubilities in the ternary MgO–CaO–MnO system, Ford and White[48] predicted a moderate solubility of MgO in CaO at 1650°C. Summarizing the work of several investigators, Troojer and Konopicky[49] predicted solubilities at 1600°C of approximately 3 mol% MgO in CaO and 1 mol% CaO in MgO.

The liquidus, solidus, and solvus lines were determined by Doman *et al.*[50] by a combination of visual and X-ray techniques. Their points are shown in Fig. 9. They reported a eutectic liquid composition of 59 mol% CaO at 2374°C. Their

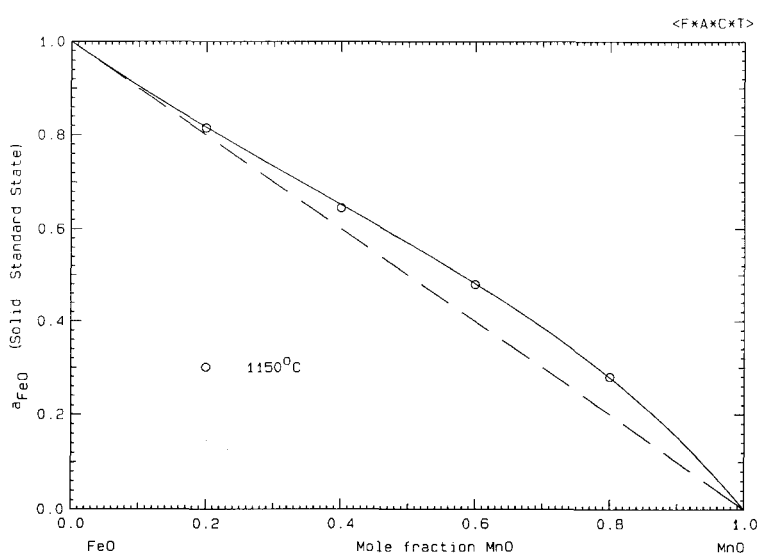

Fig. 8. Optimized activities of FeO in FeO–MnO solid solutions in equilibrium with Fe at 1150°C. Experimental points from Schwerdtfeger and Muan.[13]

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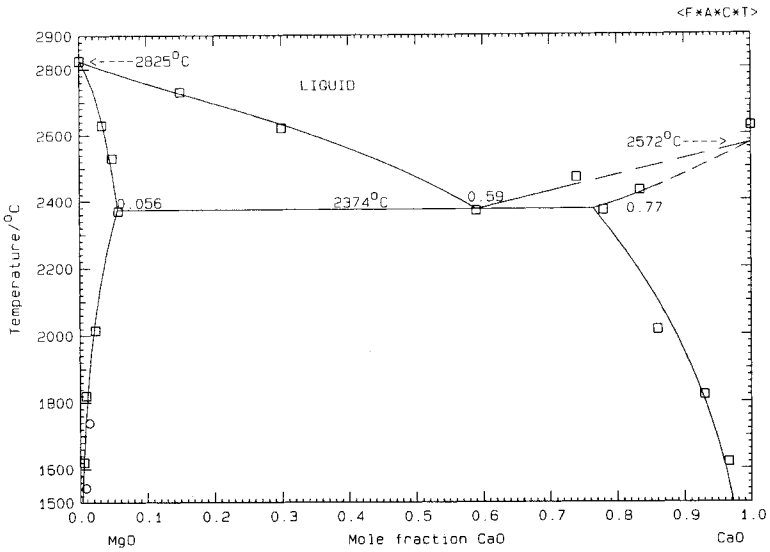

Fig. 9. Assessed MgO–CaO phase diagram. Experimental points: (□) Ref. 50, (○) Ref. 51.

reported melting point of CaO of 2630°C is 58°C higher than the present assessed value. Maximum solid solubilities at the eutectic temperature were reported as 22 mol% MgO in CaO and 5.7 mol% CaO in MgO. The solubility of CaO in MgO between 1100° and 1750°C was measured by Henney and Jones,[51] whose points are also shown in Fig. 9. Their solubilities are about twice those of Doman *et al.* The solubilities reported by Doman *et al.* were chosen in the present analysis.

The two terminal solid solutions, having the same crystal structure, were formally treated as one subregular solution with a miscibility gap with

$$G^{E}(\text{solid}) = X_{MgO}X_{CaO}(118110 - 18.410T - 31380X_{CaO}) \quad (23)$$

The liquid solution was optimized with the quasi-chemical model with

$$\omega = 45329 - 30583X_{CaO}\ \text{J/mol} \quad (24)$$

Some calculated liquid thermodynamic properties are given in Table I. As can be seen in Fig. 9, the calculated diagram agrees very well with the measured points[50] except for the liquidus at high X_{CaO} where the melting point of Doman *et al.* is higher than the assessed value.

The probable maximum inaccuracy in the assessed diagram is estimated as ±50°C for the liquidus and ±3 mol% for the solidus and solvus.

VIII. CaO–FeO System

Liquid CaO–FeO solutions in equilibrium with Fe can contain up to ≈20 wt% Fe_2O_3.[52] Nevertheless, experimental results are often represented as binary diagrams as in Fig. 10 with all Fe^{3+} converted to "FeO" with the composition axis interpreted as discussed in Section V. Compounds containing Fe^{3+} appear at lower temperatures. For example, $Ca_2Fe_2O_5$ + Fe transforms to two solid CaO–FeO solutions at 1045°C.[53] For this reason, Fig. 10 is not plotted below 1050°C.

Schuermann and Kraume[54] reported a simple eutectic diagram with a maximum solubility of CaO in solid FeO of about 16 mol%. These findings were supported by Perrot[55] and by Aubry *et al.*[56] However, Abbattista *et al.*,[53] from spectrographic measurements and microscopic observations on quenched specimens, reported a solubility of CaO in solid FeO of 37 mol% at 1100°C. In agreement with an older study by Allen and Snow[52] and a recent study by Bergman and Song,[57] they suggest that the liquidus exhibits a minimum as in Fig. 10.

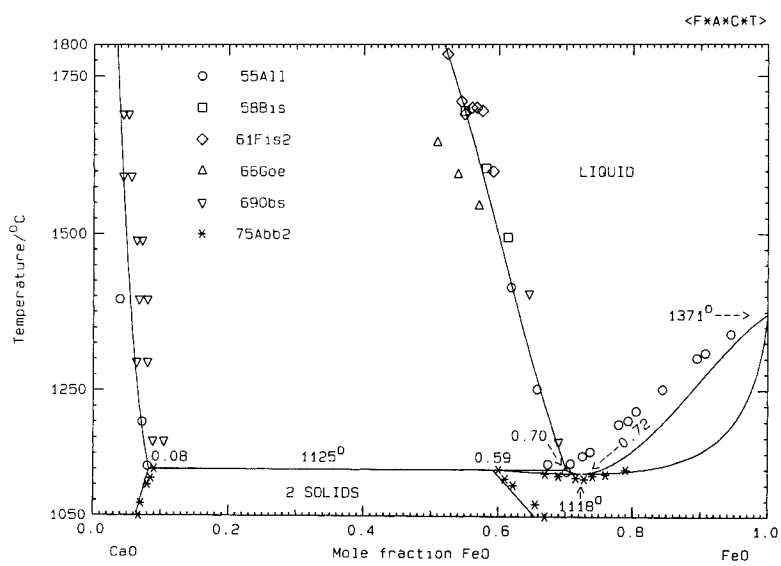

Fig. 10. Optimized CaO–FeO phase diagram for equilibrium with Fe. Experimental points: (○) Ref. 52, (*) Ref. 53, (▽) Ref. 60, (□) Ref. 63, (◇) Ref. 64, (△) Ref. 65.

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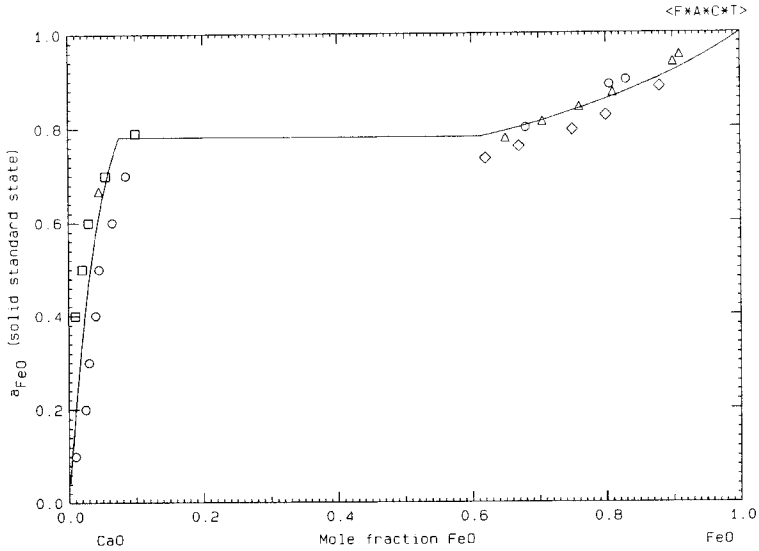

Fig. 11. Optimized FeO activities at 1100°C for CaO–FeO solid solutions in equilibrium with Fe. Experimental points: (◇) Ref. 53, (○) Ref. 69, (□) Ref. 70, (△) Ref. 71.

According to Abbattista *et al.*,[53] the reason for this disagreement may be that, in quenching experiments, the rate of cooling must be higher than 700°C per second in order to conserve the high temperature state. Bergman and Song[57] point out that the eutectic-like structure observed by Schuermann and Kraume[54] probably resulted from a eutectoid decomposition of the wustite solution into $Ca_2Fe_2O_5$ and Fe. In the present analysis, the liquidus and solvus points of Abbattista *et al.* in combination with studies of FeO solubility in CaO[52,55,59–61] and of the CaO–liquidus at higher temperatures[52,60–68] were used. Data points from several of these studies are reproduced in Fig. 10.

Activities of FeO at 1100°C in the solid solution in equilibrium with Fe were measured by several authors using gas equilibration[24,53,69] and galvanic cells with solid electrolytes.[70] As can be seen in Fig. 11, these results are in reasonable agreement, with the points of Abbattista *et al.*[53] lying below the others. In the present analysis, preference was given to the electrochemical measurements of Bjoerkman.[70]

Activities of FeO in liquid solutions in equilibrium with Fe as reported at 1600°C by Taylor and Chipman[72] and at 1400°C by Ban-ya *et al.*[73] are listed in Table II. Measurements by other authors were reviewed by Ban-ya *et al.*[73] Agreement is reasonably good among authors.

The CaO- and FeO-rich solid solutions were treated as one phase with a miscibility gap with

$$G^E = X_{CaO}X_{FeO}(22773 - 3140X_{CaO} + 15522X_{CaO}^2)\ \text{J/mol} \tag{25}$$

Good fits to the activity data (Fig. 11) and the solvus data of Abbattista *et al.* (Fig. 10) are obtained.

Four quasi-chemical coefficients were used to reproduce the liquidus and solidus and the liquid activity data:

Table II. Activity of FeO in Fe-Saturated FeO–CaO Liquid Solutions

	a_{FeO} at 1600°C		a_{FeO} at 1400°C	
X_{FeO}	exp[72]	calc	exp[73]	calc
0.60	0.43	0.44		
0.65	0.50	0.52	0.50	0.49
0.70	0.55	0.60	0.56	0.57
0.75	0.63	0.67	0.65	0.64
0.80	0.72	0.74	0.73	0.71
0.85	0.80	0.81	0.81	0.78
0.90	0.90	0.88	0.92	0.86

$$\omega = -82362 - 127477X_{FeO}^2\ \text{J/mol} \tag{26}$$

$$\eta = -20.92 - 62.76X_{FeO}^2\ \text{J/(mol·K)} \tag{27}$$

Some calculated liquid thermodynamic properties are given in Table I. The FeO–liquidus points of Allen and Snow,[52] shown in Fig. 10, are not well represented by the optimization. Moreover, the shape of their experimental liquidus is inconsistent with the existence of a minimum, which in other systems is almost always much broader, as in the optimized diagram and as suggested by the points of Abbattista *et al.* The liquidus points of Allen and Snow, measured by quenching techniques, could be in error because of a too slow cooling rate.

The liquid activity data are reproduced well as shown in Table II.

The probable maximum inaccuracy in the assessed diagram is estimated as ±2 mol% for $X_{FeO} < 0.80$ and ±50° for $X_{FeO} > 0.80$.

IX. Conclusions

Critical assessment of all available thermodynamic and phase diagram data have been carried out for the six binary systems formed among the components CaO, FeO, MgO, and MnO. All data for a system were optimized simultaneously to obtain one set of self-consistent model coefficients for the Gibbs energies of the liquid and solid solutions. These evaluations are considered to give the most reliable currently available estimates of the thermodynamic properties and phase diagrams.

In several systems, liquidus or solidus curves have not been measured. The thermodynamic analysis has permitted these to be calculated.

These optimized binary coefficients can then be used, along with appropriate models, to predict the thermodynamic properties of ternary and multicomponent systems. We are currently evaluating and optimizing data on all subsystems of the SiO_2–Al_2O_3–CaO–MgO–FeO–MnO–TiO_2–Ti_2O_3–ZrO_2–Na_2O–K_2O–S system.

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